

Algebraic Equations

Name: _____

Add all the like terms

Part A

Eg: $3a + 4b + 6a + 3b$
 $= 3a + 6a + 4b + 3b$
 $= 9a + 7b$

① $6n + 7m + 8n + 4m$

② $y + 3e + 5y + 2e$

③ $8x - 7p - 6x - 3p$

④ $6q + 12r + 3q - 11r$

Expand the brackets

Part B

Eg: $3(2f + 4)$
 $= 6f + 12$

① $2(2e + 5)$

② $3(4t + 2)$

③ $4(3r - 5)$

④ $6p(5 - 2p)$

Brackets and like terms

Part C

$$\text{Eg: } 3(4+x-2x^2) + 2x(2x+1)$$

$$= 12 + 3x - 6x^2 + 4x^2 + 2x$$

$$= 12 + 3x + 2x - 6x^2 + 4x^2$$

$$= 12 + 5x - 2x^2$$

$$\textcircled{1} 2(n^2 + 3n + 4) + n(5n + 2)$$

$$\textcircled{2} g(g+3) + 3(2g^2 + 4g + 1)$$

Solve the following equations

Part D

$$\textcircled{1} 2x = 32$$

$$\textcircled{2} 7p = 42$$

$$\textcircled{3} 3n = 36$$

$$\textcircled{4} x + 4 = 9$$

$$\textcircled{5} u - 6 = 2$$

$$\textcircled{6} h + 4 = 3$$

$$\textcircled{7} 3x + 1 = 13$$

$$\textcircled{8} 2x + 1 = 9$$

$$\textcircled{9} 2x + 4 = 10$$

Solve the following equations

Part E

Hint: like terms!

① $6a + 5a = -11$

② $-6n - 2n = 16$

③ $4x + 6 + 3 = 17$

④ $3(2x + 9) = 39$

⑤ $2(x + 1) = 14$

⑥ $4x + 3 = 2x + 11$

⑦ $8(x + 1) = 2(x + 16)$

⑧ $4(2x - 1) = 3(x + 2)$